

Role Allocation for Hub-Delivered Knowledge in Multi-Robot Exploration

A matched-tuning comparison of explorer–messenger specialization
and homogeneous return-to-hub swarms

Giovanni Mottini

Matricola 933100

`g.mottini1@campus.unimib.it`

Giacomo Maggioni

Matricola 869265

`g.maggioni24@campus.unimib.it`

25 July 2026

Abstract

An exploration team has two jobs: build the map, and make it available. In most missions the second job is what counts, because distributed knowledge is not directly usable until it reaches a base station. This report asks how a fixed team of mobile robots should divide itself between the two, and weighs two answers against each other. EXMSG splits the team into sensing explorers and blind messengers whose only task is carrying knowledge to a central hub. Unified keeps the team homogeneous: every robot senses, and every robot walks its own map back to deliver the knowledge. That comparison is the subject of the report. Both run in the same grid simulator, and in both, progress is measured only by what the hub knows. Our first campaign tunes EXMSG against fixed reference designs. That stacks the deck in favor of the tuned line, so a second campaign gives unified an equal tuning budget and the same limited communication range, leaving role allocation as the only difference between the two architectures. An idealized stationary-relay team and a free-communication variant of the swarm serve as untuned upper references. Validation on 21 fixed maps from seven topology families splits the answer in two. EXMSG finishes every map; every homogeneous variant, idealized communication included, fails the two large mazes, and on those two layouts nothing finishes without dedicated carriers. On the maps both teams do finish, though, the tuned swarm is slightly faster overall and wins five of the seven topology families. EXMSG keeps a clear speed advantage only on corridor- and maze-like layouts; elsewhere what it has left is a lower fleet cost. Specialization, in short, buys completion where delivery travel is expensive and trades speed for cost everywhere else. We assume known poses, noise-free sensing, and oracle navigation: this is not multi-robot SLAM.

Keywords: multi-robot exploration; role allocation; heterogeneous robot teams; mobile relays; communication constraints; frontier exploration; matched-baseline tuning

1 Introduction

1.1 Motivation

Sending a team into a large or hazardous environment beats sending one robot, because the whole mission no longer rests on a single machine. Search-and-rescue reconnaissance and the inspection of subterranean or otherwise communication-limited spaces are the standard examples. Spreading sensing across a team can speed up coverage, but only if the robots avoid redoing each other’s work and can make their observations available to the rest of the mission (Yamauchi, 1998; Burgard et al., 2005).

That last task carries most of the difficulty. A robot may have covered a whole floor while out of contact, yet no operator and no base station can use what it found until the map is delivered. Prior work keeps the team connected, deploys relays, or arranges intermittent reconnection (Rooker and Birk, 2007; Amigoni et al., 2017). What these approaches share is an awareness of something that plain explored-area metrics miss: *discovering* information and *delivering* it are separate processes, and both spend mission time.

1.2 Research gap and question

The design question is therefore not only where robots should explore, but who does the carrying. A homogeneous swarm keeps every robot sensing and pays for it each time one breaks off to walk its map home. A heterogeneous team keeps its explorers in the field by reserving some robots for delivery, and gives up their sensing capacity in exchange. Stationary relays are a third option: infrastructure dropped once and then left in place, which pays off or not depending on where it lands and on what the network is assumed to do.

These designs are usually studied under different objectives and different communication models, which makes any measured difference between them hard to attribute: architecture design, or parameter settings? We therefore run all of them in one simulator under a single rule. A map cell counts toward mission progress only once its observation has reached the hub’s knowledge map, `hub_pool`. What the report studies is the pairing itself: EXMSG, the explorer–messenger design, against unified, the homogeneous one. A first campaign tunes EXMSG against fixed, untuned reference designs. A second campaign then hands unified an equal tuning budget and the same communication constraints, so the comparison turns on role allocation only; the idealized stationary-relay design is left alone as an untuned ceiling. The primary research question is:

Under a hub-delivered knowledge objective, when does allocating part of a mobile team to dedicated knowledge delivery beat a homogeneous swarm in which every robot senses and delivers, and how does the answer depend on map topology and robot cost?

The scope stops well short of multi-robot SLAM. Robots share one coordinate frame, and every campaign reported here uses known-pose grids and oracle navigation.

1.3 Study design and contributions

The project makes four contributions:

1. **A delivery-centered experimental framing.** Hub coverage draws a line between knowledge scattered across the robots and knowledge the mission can act on, and draws it without demanding continuous connectivity.

2. **A matched-tuning evaluation design.** EXMSG first meets two deliberately untuned references: homogeneous return-to-hub exploration and an idealized stationary relay. The second campaign then removes the obvious objection to that arrangement. The homogeneous line gets its own sequential tuning budget (return policy first, then coordination) and the same range-limited communication EXMSG uses, which leaves role structure as the only difference between the two. The idealized references stay untuned ceilings and take no part in cost claims.
3. **Two complementary experimental campaigns.** A five-phase, 7,450-run campaign tunes EXMSG’s role allocation, coordination, and communication, then validates it on 21 fixed maps from seven topology families at three scales. A second campaign of 7,326 runs (Phases 6–9) retunes the homogeneous baseline, re-checks EXMSG’s composition, and revalidates every line together.
4. **A reusable interactive simulation platform.** A single Python engine drives both the headless batch runner behind the campaigns and a browser interface with live visualization, side-by-side simulation slots, and parameter-sweep tooling (Appendix A).

1.4 Report organization

Section 2 places the work among coordinated frontier exploration, planning under communication limits, relay systems, and multi-robot mapping. Section 3 defines the hub objective, the architectures, and what the simulator does and does not model. The campaigns and their results follow. Section 6 reads those results against the literature and marks the point where they stop being valid. Appendices A and B document the simulator built for this study and its full tunable-parameter catalogue.

2 Related Work

The project sits where three lines of work meet: multi-robot frontier exploration, coordination under communication limits, and relay-mediated data delivery. None of them offers an interchangeable benchmark for what we do here, so each subsection below states what role that line plays in this study and which assumptions keep it from being directly comparable. Table 1 collects the comparison.

2.1 Frontier-based and coordinated exploration

Frontiers mark the boundary between known free space and unexplored space (Yamauchi, 1997). Handing the method to several robots raises the coverage rate, and also produces redundant assignments as soon as two robots pick the same region (Yamauchi, 1998). Coordinated approaches answer that by combining travel cost with expected frontier utility and discounting targets that teammates have already claimed (Burgard et al., 2000, 2005).

Our simulator keeps the frontier abstraction but coordinates with two lighter mechanisms: repulsion, which penalizes candidates near another explorer’s target, and reservation, which withdraws claimed targets for a time. Neither solves a global assignment problem. The literature gives good reason to measure duplicate targets and travel efficiency; it does not promise that strict reservation minimizes completion time.

2.2 Communication-restricted exploration

Work on communication-restricted exploration divides roughly into keeping the team continuously connected, reconnecting intermittently, and deploying relays (Amigoni et al., 2017). Rooker and Birk explore under a wireless connectivity constraint tied to a fixed base station (Rooker and Birk, 2007). CBAX goes further and treats frontier selection, relay placement, routing, connectivity, and link capacity as one joint problem (Pei et al., 2010). In all of these, communication decides which motions or assignments are feasible; it is not a post-processing concern.

We take the same premise and apply a different constraint. Robots may disconnect, and progress is scored only once information reaches the hub. That measures the cost of delayed connectivity without insisting on a connected network at every step. There are no network flows, queues, contention, or routing anywhere in the model.

2.3 Role specialization, rendezvous, and mobile relays

Role-based exploration separates explorer duties from relay duties (de Hoog et al., 2009), an attractive split whenever sensing progress and communication support compete for the same robot-minutes. Related data-gathering formulations put source robots to work generating information and relay robots to work uploading it to a data center, under buffer and intermittent-communication constraints (Guo and Zavlanos, 2018). Rendezvous-based exploration takes another route and plans explicit meetings where disconnected robots trade map knowledge (da Silva et al., 2024).

EXMSG belongs to this family: explorers generate occupancy information, messengers handle pickup and hub delivery. The resemblance is architectural, not algorithmic. Our messengers pick targets heuristically, carry unbounded map knowledge, follow no optimized schedule, and navigate with oracle knowledge. EXMSG is a stripped-down source-relay abstraction, not an implementation of any planner cited above.

2.4 Stationary relays and low-bandwidth coordination

Relay-placement work asks where communication support should be created as a team advances; CARA, for instance, assigns relay robots according to communication context during an expedition (Abu-Aisheh et al., 2022). At the other end of the bandwidth spectrum, decentralized exploration can skip dense map exchange altogether and share something compact instead, such as robot positions (Bayer and Faigl, 2026). Read together, the two show that relay topology and message representation are independent design choices.

EXANCH idealizes the first of them. Explorers deploy stationary anchors along a connected geometric frontier, but every anchor shares one knowledge array, so information reaches the hub with no forwarding, hop delay, internal bandwidth, or routing failure. Our bandwidth stress works on occupancy cells too, not on the position-only messages of low-bandwidth coordination work.

2.5 Mapping uncertainty and the SLAM boundary

Occupancy-grid mapping represents spatial belief probabilistically and revises it as uncertain observations accumulate (Elfes, 1989). Active-mapping research couples that estimation to sensing, planning, localization, and information gain (Lluvia et al., 2021). Full multi-robot SLAM asks for more still: relative localization, frame alignment, loop closure, consistency, and map fusion (Saeedi et al., 2016; Yu et al., 2020; Kegeleirs et al., 2021).

We stop deliberately before that boundary. The simulator’s belief mode counts votes rather than calibrated probabilities, both campaigns run at zero sensing noise, and all maps share one coordinate frame, so an exchange never has to register anything. What we study is map-knowledge delivery on a known-pose grid, not multi-robot SLAM.

2.6 Position of the present study

The question running through all of these lines is how communication enters the exploration problem. What we add is a shared hub-delivered objective and a head-to-head comparison of two ways to spend one mobile budget: a source-relay team, EXMSG, against a homogeneous return-to-base swarm, both in the same simulator, with an idealized stationary-relay design as a ceiling above them. Since tuned-vs-untuned fairness is a recurring worry in this literature, the homogeneous control also receives a matched tuning budget and matched communication constraints in a second campaign. We propose no new frontier allocator, rendezvous planner, routing protocol, or SLAM back end.

Table 1: Literature-to-system comparison. Alignments motivate the experimental questions; differences bound the conclusions.

Research line	Project analogue	Useful alignment	Critical difference
Coordinated frontiers	Repulsion and reservation	Reduces duplicate targets	Heuristic rather than optimized assignment
Connectivity and bandwidth	Hub objective; range and cell cap	Delivery is mission-relevant	Disconnection is permitted; no routing model
Mobile data relays	EXMSG	Sensing and delivery roles are separated	No finite buffers or scheduled rendezvous
Relay placement	EXANCH	Connected infrastructure supports delivery	Shared pool replaces multi-hop forwarding
Low-bandwidth coordination	Cell-transfer cap	Limits information per exchange	Still exchanges map cells rather than compact positions
Probabilistic mapping	Belief mode	Repeated evidence can revise cells	Vote counts are not calibrated occupancy probabilities
Multi-robot SLAM	Shared local maps	Information is exchanged	No relative pose or map registration

3 System Model and Methods

3.1 Environment and mission objective

The environment is a finite two-dimensional occupancy grid. Every cell holds one ground-truth state, free or obstacle, while a robot’s knowledge map holds one of three values: unknown, free, or obstacle. Fixed maps come from ASCII layouts. We seal their outer borders and keep only the largest four-connected free-space component, which guarantees that every free cell in the evaluation is reachable from wherever the robots start.

The hub is the free cell nearest the geometric center of the map, and every mobile robot starts on it. Let $\mathcal{P}$ be the set of ground-truth free cells, with $P = |\mathcal{P}|$. The simulator keeps a separate

hub knowledge map, `hub_pool`. With $K_h(t)$ the number of cells in $\mathcal{P}$ that the hub knows to be free after step t , hub coverage is

$$C_h(t) = 100 \frac{K_h(t)}{P}. \quad (1)$$

The mission is complete only when $K_h(t) = P$. A cell an explorer has discovered does not contribute to the evaluation until that knowledge reaches the hub through whatever delivery mechanism its architecture provides. Every architecture is scored on the same thing: how fast it closes that gap between local discovery and hub delivery.

3.2 Sensing and knowledge representation

Only explorers sense. At each step an explorer observes the cells inside a configured sensing radius and a heading-centered cone; both campaigns fix these at five grid cells and 120° in every configuration. A Bresenham line-of-sight test drops cells hidden behind obstacles. When an explorer’s planned path runs out, it senses toward each adjacent unknown direction before replanning. At a frontier this corner peek exposes free space around sharp turns that a forward-facing cone would miss.

Every robot carries a local knowledge map. Under the locked knowledge model, an arriving value, observed or received, fills an unknown cell and never overwrites a known one. The simulator also implements a vote-based belief model, but neither campaign uses it. Every run reported here sets both false-empty and false-obstacle probabilities to zero, so the locked rule introduces no sensing errors in these experiments.

Alongside the robot-local maps, the simulator keeps an aggregate explorer map for visualization and measurement. In EXMSG and EXANCH that aggregate feeds nothing back to the explorers: their local maps change only through sensing or through the relay interaction their architecture permits. The hub map is separate from both and stays the authority for coverage and termination.

3.3 Robot architectures

EXMSG and unified are the two architectures the report weighs against each other; EXANCH sits above both as a reference ceiling. The hub-delivery campaign holds unified and EXANCH fixed as untuned controls, and the matched-tuning campaign then tunes unified while EXANCH stays an untuned ceiling. Let N be the mobile-robot budget. Unified and EXANCH spend it entirely on explorers, EXMSG splits it between explorers and messengers by a configurable ratio, and EXANCH adds a map-dependent anchor inventory on top. Both campaigns fix $N = 10$ for the primary comparison; Phase 8 also sweeps N to build an equal-cost control.

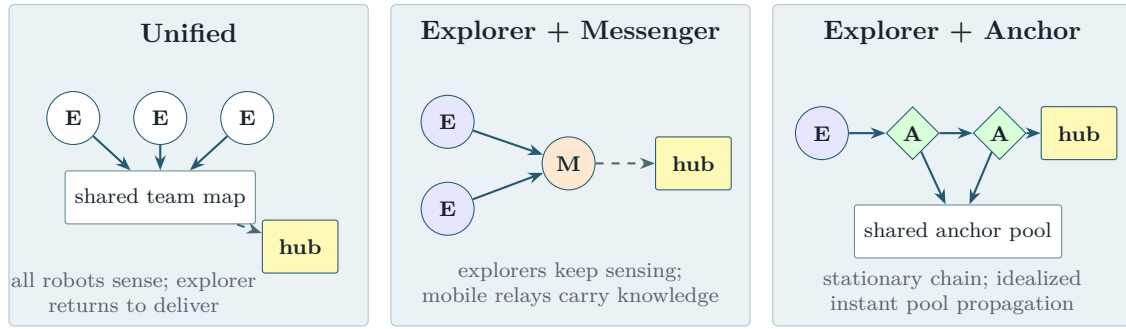


Figure 1: Schematic information flow in the two compared architectures, homogeneous unified (left) and EXMSG (center), with the idealized stationary-relay reference EXANCH (right). The number of icons does not represent fleet size. Solid arrows show knowledge transfer; dashed arrows show physical return travel. E, M, and A denote explorer, messenger, and anchor robots, respectively.

Explorer + Messenger (EXMSG, first compared design). EXMSG divides the N mobile robots into sensing explorers and blind messengers, with the explorer ratio setting the split and at least one explorer guaranteed. The premise is an explicit division of labor: explorers never break off frontier work to deliver, and messengers absorb every hub-delivery trip. Explorer-explorer exchange is switched off. A seeking messenger heads for the nearest explorer, with a virtual distance penalty that discourages two messengers from converging on the same one; knowledge moves between explorer-messenger and messenger-messenger pairs within communication range. A messenger holding undelivered knowledge switches to delivery only after a physical meeting within 1.5 cells and, end-game safeguard aside, outside a two-cell hub deadzone. It then drops its payload within one cell of the hub and goes back to seeking. Messenger speed is counted in grid moves per step.

The other two designs sit on opposite sides of it: unified removes delivery specialization altogether, EXANCH replaces mobile delivery with idealized stationary infrastructure.

Unified (homogeneous, second compared design). Unified makes all N mobile robots explorers, with no dedicated relays, and comes in two communication variants. The *pool* variant shares a global team map every step and acts as an idealized free-communication ceiling. The *local* variant confines explorers to the same proximity-based communication range EXMSG uses (Section 3.5), so the two compared lines share one communication mechanism and differ only in role structure. Delivery is physical in both: an explorer has to stand on the hub cell to deposit its map. Three parameters govern the return: an eligibility timer (`hubReturnInterval`), an idle threshold (`hubReturnIdle`), and a hard cap (`hubReturnMax`), fixed at 40, 10, and 90 steps in the hub-delivery campaign. Phase 6 caught the 90-step cap pulling explorers home before they could reach distant frontiers on large maps, and retuned it to 360. Unified is what isolates the cost EXMSG is designed to remove: interrupting exploration to deliver.

Explorer + Anchor (EXANCH, idealized stationary-relay reference). EXANCH also makes all N mobile robots explorers and adds a density-derived inventory of stationary anchors, one of which starts at the hub. An explorer may drop another on an explored cell lying within anchor communication range of the existing network. Explorers exchange knowledge only with nearby anchors. All anchor instances point at the same knowledge array, so whatever is uploaded to any anchor is instantly available at the hub: no hop-by-hop messages, no delay, no loss, no bandwidth spent inside the chain. EXANCH is an optimistic upper bound for connected relay infrastructure, not a model of a physical network.

3.4 Frontier exploration and navigation

A frontier is a known free cell with at least one four-connected unknown neighbor. Explorers plan shortest paths by breadth-first search (BFS). Under the nearest policy that the final configurations use, each replan looks at the 20 geometrically nearest frontiers plus up to ten randomly sampled farther ones, and takes the candidate with the lowest BFS path cost.

The implementation also rejects a frontier candidate whose ground-truth cell is an obstacle. Across the reported campaigns this filter is redundant, since sensing noise is zero, but in a noisy campaign it would hand the explorer an oracle advantage, so we treat it as a model-validity limitation.

Repulsion adds cost to a candidate lying near another explorer’s current target. Inside radius r_R the penalty grows with proximity and is capped at twice the candidate’s path cost. Reservation, when enabled, instead excludes frontiers another explorer has claimed; claims expire after 30 steps or when the owner changes target. If no normal frontier path is available, the explorer searches the full map for the nearest reachable unknown cell, then falls back to the hub or to a heading-biased walk.

The simulator supports oracle, known-free, and risk-unknown navigation. Every reported run uses oracle navigation: BFS tests motion against the ground-truth grid even where that geometry is missing from the robot’s own map. This takes navigation failure out of the comparison, at a real cost in realism, most of all for blind messengers.

3.5 Communication model

Except for the unified pool variant and the idealized anchor backbone, robot-to-robot exchange is proximity based. A pair is eligible when its Euclidean grid distance falls within the configured range. A transfer can be limited to B cells per direction and dropped independently with probability p_{loss} . When $B > 0$, the campaign’s newest-first policy orders cells by their local knowledge timestamp. Distance- and line-of-sight-dependent link quality are implemented but unused by either campaign. The final EXMSG line runs at range 10, unlimited transfer size ($B = 0$), and zero packet loss, while Phase 4 varies all three; the matched-tuning campaign’s local-unified line inherits the same range-10 constraint as a shared campaign-wide radio assumption.

We keep two communication counters apart. An attempt is any eligible simulated exchange, including one that moves no new cells; a successful knowledge event changes at least one map. The unified local variant counts range-limited pairwise exchanges exactly as EXMSG does. Attempts per 100 steps measure interaction activity under each simulator abstraction and should not be read as radio packets, bytes, or energy.

3.6 Outcome measures

The primary outcomes are completion rate and completion steps. A configuration can fail to finish within the horizon, so mean and median completion steps are computed over completed runs only and always reported next to completion rate. All-run step summaries keep the horizon value for runs that exhaust it and the observed terminal step otherwise.

Coverage speed is summarized over a fixed phase horizon T :

$$\text{AUC}_T = \frac{1}{T} \sum_{t=1}^T C_h(t). \quad (2)$$

If a run ends before T , its terminal coverage is carried forward to the horizon, so early completion

stays at 100% and early non-completion stays on its terminal plateau. The score is in percentage points and comparable only across runs that share the same T .

The fleet-cost proxy is

$$J_{\text{fleet}} = c_E (n_E + r_M n_M + r_A n_A), \quad (3)$$

with $c_E = 3$, $r_M = 0.33$, and $r_A = 0$ in the campaign. For EXANCH, n_A is the planned anchor inventory, not merely the anchors actually deployed. These weights are configurable relative costs; they measure neither money nor energy.

The secondary outcomes are attempts and knowledge-changing events per 100 steps, newly delivered hub cells per step, messenger pickup and delivery latencies, times to 50%, 80%, and 95% hub coverage, explorer replans and BFS failures, and duplicate frontier targeting. The duplicate-target rate is the percentage of executed steps in which at least two active explorers share a target.

4 Experimental Methodology

4.1 Sequential campaign design

The hub-delivery campaign runs in five sequential phases. Phase 1 enestablishes a baseline for each architecture. Phases 2–4 optimize EXMSG’s role allocation, coordination, and communication in that order. Phase 5 validates the resulting EXMSG line against the unchanged unified and EXANCH references. Table 2 summarizes the campaign design. The references are left untuned on purpose: they enter Phase 1 and Phase 5 in fixed reference configurations.

Table 2: The hub-delivery campaign. “Runs” counts all parameter combinations, maps, and repeated seeds.

Phase	Design	Runs	Decision
1	3 architectures $\times$ 4 core maps $\times$ 30 seeds	360	Establish reference behavior
2	5 explorer ratios $\times$ 4 messenger speeds $\times$ 2 frontier policies $\times$ 4 maps $\times$ 20 seeds	3,200	Select EXMSG role allocation
3	2 reservation states $\times$ 2 repulsion states $\times$ 2 radii $\times$ 4 maps $\times$ 20 seeds	640	Select EXMSG coordination
4	Range sweep plus bandwidth–loss matrix on 4 maps and 20 seeds	1,360	Select EXMSG communication settings
5	3 finalist lines $\times$ 21 maps $\times$ 30 seeds	1,890	Validate across families and scales

4.2 Matched-tuning campaign

Phase 5 compares a fully tuned EXMSG line against references from the Phase 1 configurations. Unified receives a tuning budget matched to EXMSG’s own sequential procedure, return policy first and then coordination, and is held to the same range-limited local communication EXMSG uses (`unifiedComm=local`); EXANCH remains an untuned ceiling throughout. The new phases continue the first campaign’s numbering as Phases 6–9, and Table 3 gives the factorial scope.

Table 3: The matched-tuning campaign. A 16-run preflight and an interactive diagnostic preceded Phase 6; both are described in the text and omitted from the run tally here.

Phase	Design	Runs	Decision
6	3 return intervals $\times$ 3 idle thresholds $\times$ 3 hard caps $\times$ 4 core maps $\times$ 20 seeds	2,160	Select unified return policy jointly
7	2 frontier policies $\times$ 2 repulsion radii $\times$ 2 reservation states $\times$ 4 core maps $\times$ 20 seeds	640	Select unified coordination
8	Unified $N \in \{6, 7, 8, 10, 12\}$ and EXMSG per- N composition sweep, each $\times$ 4 core maps $\times$ 20 seeds	1,360	Re-check EXMSG composition at range 10; build cost frontiers
9	5 finalist lines $\times$ 21 maps $\times$ 30 seeds	3,150	Validate all lines under matched tuning

Phase 6 began by diagnosing why unified had plateaued on large, low-connectivity maps in the hub-delivery campaign. An interactive inspection and a paired headless counterfactual caught the 90-step return hard cap recalling explorers before they could reach distant frontiers, while the frontier and delivery mechanisms themselves worked as intended. Phase 6 then tuned the return interval, idle threshold, and hard cap jointly, settling on 40, 10, and 360 (Section 3). Phase 7 tuned coordination on top of that policy and chose the nearest frontier policy, repulsion radius 8, and reservation disabled: the same coordination settings the hub-delivery campaign had selected for EXMSG, arrived at independently. Phase 8 then re-checked EXMSG’s own role composition at the operational communication range (10 cells, selected in Phase 4). Six explorers and four messengers, the hub-delivery campaign’s original choice, did not pass that re-check. Seven explorers plus three messengers won instead (Section 5) and became the finalist used in Phase 9 and throughout the remainder of this report. Unified at $N = 8$ is that finalist’s equal-cost twin under the campaign’s messenger cost ratio, while unified at $N = 10$ remains the equal-mobile-robot-count comparison. Phase 9 then validates five lines together: EXMSG, tuned local unified at $N = 10$ and $N = 8$, an idealized free-communication ceiling for unified (`unifiedComm=pool`, same tuned return policy and coordination), and EXANCH.

4.3 Maps and controlled factors

Phases 1–4 run on four standard-scale core maps: rooms, corridors, bottlenecks, and labyrinth. Final validation widens to seven families at small, standard, and large scale, giving 21 fixed environments. Table 4 lists dimensions and free-cell counts after the connected-component filter. Each family stresses something different. Rooms test doorway transitions, corridors long constrained routes, bottlenecks chokepoints between wider regions, and labyrinths dense turns and narrow passages; loops add alternative routes, columns add clutter to otherwise open space, and open maps represent empty space.

Table 4: Fixed-map catalogue used in final validation. Each scale entry is width $\times$ height / free cells.

Family	Small	Standard	Large
Rooms	30 $\times$ 30 / 577	51 $\times$ 51 / 1,756	70 $\times$ 65 / 3,040
Corridors	30 $\times$ 30 / 436	50 $\times$ 49 / 1,211	71 $\times$ 62 / 2,163
Bottlenecks	30 $\times$ 30 / 598	50 $\times$ 49 / 1,768	70 $\times$ 68 / 3,552
Labyrinth	31 $\times$ 31 / 449	51 $\times$ 51 / 1,249	61 $\times$ 61 / 1,799
Loops	30 $\times$ 30 / 537	51 $\times$ 50 / 1,696	72 $\times$ 62 / 3,119
Columns	30 $\times$ 30 / 700	50 $\times$ 50 / 2,064	70 $\times$ 67 / 4,066
Open	30 $\times$ 30 / 784	50 $\times$ 50 / 2,304	70 $\times$ 70 / 4,624

Figure 2 shows the standard-scale geometry used throughout the tuning phases and as the middle scale in final validation.

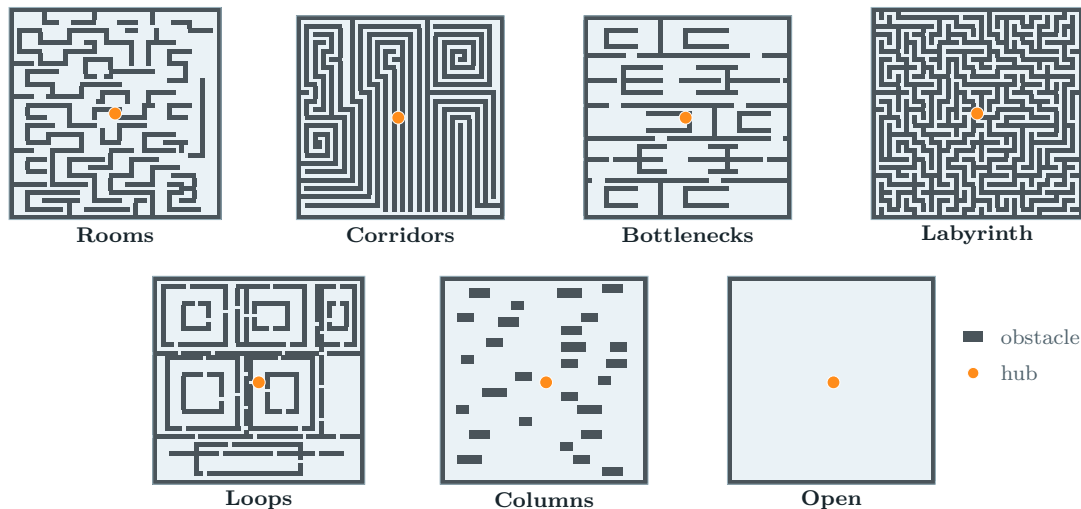


Figure 2: Standard-scale fixed maps. Dark cells are obstacles, pale cells belong to the connected free-space component used by the simulator, and the gold marker shows the hub at the nearest free cell to the map centre. All thumbnails use the same cell scale.

Across both campaigns and every configuration, the primary mobile-robot budget is fixed at $N = 10$; the matched-tuning campaign additionally sweeps N in Phase 8 to build an equal-cost control. Unless a phase sweeps them, runs use sensing range 5, cone angle 120° , nearest explorer frontiers, nearest-explorer messenger targeting, oracle navigation, zero sensing noise, and no scheduled failures. All mobile robots spawn at the same hub, and EXANCH starts with one anchor there. The hub-delivery campaign’s Phase 5 EXMSG configuration split the mobile budget into six explorers and four messengers; the matched-tuning campaign’s Phase 8 re-check, run at the operational communication range, picked seven explorers and three messengers instead, and that is the EXMSG finalist used in the matched-tuning campaign’s final validation (Section 5). Both finalists use messenger speed 4, repulsion radius 8, communication range 10, unlimited bandwidth, and zero packet loss. The unified finalist shares the repulsion radius and communication range, with reservation disabled and a return policy of interval 40, idle 10, and hard cap 360. The final EXANCH reference uses ten mobile explorers plus a map-dependent anchor inventory derived from density 0.015, with anchor range 6. What the comparison controls is the number of mobile robots, not the total number of robotic units. Appendix B catalogues every tunable parameter and the default values.

4.4 Seeds, horizons, and execution

Every parameter combination inside a phase gets the same contiguous seed set, which is what makes paired-seed comparisons possible, and every phase across both campaigns draws on an independent seed block. Appendix A lists the specific ranges in its reproducibility notes. Randomness touches initial headings, sampled frontier candidates, fallback movement, and any stochastic communication or sensing setting a phase turns on.

Tuning phases in both campaigns run to an 8,000-step horizon and record coverage every 100 steps; both final validations run to 12,000 steps with a 50-step curve interval. A run ends early on full hub coverage or on a terminal failure condition, and is otherwise recorded as non-complete at the horizon. Coverage milestones that are never reached stay missing rather than being imputed as successful times.

Both campaigns ran under Python 3.12.13 with two process workers. Each campaign’s own preflight confirmed identical outcomes and seed-level step counts with one and two workers before the full experiments started. Every output directory holds the resolved manifest, environment metadata, per-run CSV, and grouped summary CSV; the matched-tuning campaign also writes one dated decision entry per phase to a decisions log, recording the selected configuration, the ranking rule, and the runner-up gap.

Every recorded run executes headless, but the artifact underneath is a complete interactive simulator. The same engine serves a browser frontend over WebSocket, with live map visualization, two independent simulation slots for side-by-side configuration comparison, and an in-browser parameter-sweep panel. Because the headless batch runner and the interface share that one engine, what we saw interactively and what the experiments recorded cannot drift apart. The interface is where behavior inspection and hypothesis generation happened while the campaign phases were being designed.

4.5 Selection and statistical summaries

Automated summaries rank configurations by completion rate, then break ties on median completion steps, coverage AUC, mean completion steps, fleet cost, and communication attempts per 100 steps, in that order, preferring the better value at each stage. Phase 2 selects on the mean of map-level medians rather than a pooled seed median, so a majority of easy maps cannot hide losses on corridors or labyrinths. Phase 3 deliberately treats duplicate targeting and speed as separate objectives. Phase 4 checks for outright failure and reports observed degradation points before locking the final communication settings. Phases 6 and 7 apply that same rule to that same mean-of-map-level-medians aggregate, for the same reason; ties there go to whichever setting sits closest to the archived hub-delivery-campaign defaults. Phase 8’s composition re-check was fixed in advance: it re-tests compositions at the operational communication range instead of opening a new search. Phase 9 reports completion first and treats speed as conditional on it, which is what produces the two-tier reliability/cost structure of Section 5 rather than one pooled ranking across all five lines.

For each configuration–map group, the runner reports completion rate, completed-run mean and median steps, population standard deviation, and a normal-approximation 95% confidence interval for the mean, along with all-run step summaries and coverage-milestone reach rates. The study is descriptive. Paired seeds cut comparison noise, but we run no hypothesis tests, and the maps are not a random sample of environments. Conclusions are bounded to the implemented simulator, the fixed map catalogue, the parameter ranges, and the sequential selection procedure.

5 Results

Sections 5.1–5.4 follow the hub-delivery campaign’s own EXMSG tuning history (Phases 1–4). Section 5.5 reports the matched-tuning final campaign’s Phase 9 validation, which supersedes the hub-delivery campaign’s untuned-reference Phase 5 as the report’s final result. The final coverage curves are regenerated from the 3,150 Phase 9 run records, with terminal coverage carried forward to the common 12,000-step horizon exactly as in the AUC definition.

5.1 Phase 1: untuned reference behavior

All 360 Phase 1 jobs ran without a runtime failure. EXMSG, still in its untuned Phase 1 configuration, completed every seed on all four core maps, and so did the idealized relay reference.

Unified completed every rooms and bottlenecks run, and not a single corridors or labyrinth run within 8,000 steps. Table 5 therefore fixes the reference gap before any tuning happens: the homogeneous baseline was fastest on rooms and failed both constrained topologies outright, while EXMSG kept complete coverage everywhere.

Table 5: Phase 1 core-map results recomputed from the grouped campaign output. Completion entries give completed runs as a percentage; UNI, E+M, and E+A denote Unified, EXMSG, and EXANCH.

Map	UNI	E+M	E+A	Fastest completed median	Selected AUC evidence
Rooms	100	100	100	Unified: 250	EXMSG 98.6; EXANCH 98.6
Corridors	0	100	100	EXMSG: 768	Unified 52.3; EXMSG 95.0
Bottlenecks	100	100	100	EXMSG: 340	EXMSG 98.2
Labyrinth	0	100	100	EXANCH: 941; EXMSG: 1,057.5	Unified 29.9; EXMSG 94.0; EXANCH 95.7

The two incomplete unified cases also produced the lowest AUC values on record, 52.3 on corridors and 29.9 on labyrinth. EXMSG stayed between 94.0 and 98.6 across all four maps, and EXANCH led the labyrinth comparison at 95.7. EXMSG reached that full core-map completion with a fleet-cost proxy of 21.96 against 30.0 for each reference, and with 426.5 to 556.1 communication attempts per 100 steps against unified’s roughly 2,005.

5.2 Phase 2: EXMSG role allocation

All 3,200 Phase 2 jobs completed, which left the sweep with no distinct reliability-oriented configuration to identify, so selection fell to map-level completion speed. The winning line used an explorer ratio of 0.6, messenger speed 4, and the nearest frontier policy, for a mean of map-level median steps of 549.2.

Table 6: Map-level behavior of the selected Phase 2 EXMSG configuration.

Map	Median steps	Delivery latency	Delivered cells / step
Rooms	269.0	5.95	9.67
Bottlenecks	274.5	6.77	8.80
Corridors	684.5	9.46	3.53
Labyrinth	969.0	14.93	2.69

The topology gradient shows up in both delivery measures: corridors and labyrinth raised messenger latency and cut the rate at which new hub cells arrived. The selected 0.6/4 configuration beat the closest alternative (0.7 explorer ratio, speed 3) on every hard map, even though that alternative had the lower pooled seed-level median. We kept the map-level criterion precisely so that the easy rooms and bottlenecks cases could not dominate selection.

This composition, six explorers and four messengers, carries through the rest of the hub-delivery campaign, Phase 5 validation included. It does not survive re-examination once the communication range is pinned at Phase 4’s own selection of 10 cells: the matched-tuning campaign’s Phase 8 re-check at $N = 10$ under range 10 picks seven explorers and three messengers instead (Section 4.2), and that is the composition used in the final validation reported in Section 5.5.

5.3 Phase 3: EXMSG coordination

Phase 3 completed all 640 jobs and recorded no explorer BFS failures in either principal configuration. Reservation eliminated duplicate frontier targets without minimizing completion time. The speed-first line disabled reservation, enabled repulsion, and raised its radius to 8 (Table 7).

Table 7: Phase 3 coordination trade-off. Step and AUC values are means of the four map-level grouped summaries.

Configuration	Median steps	AUC	Duplicate-target steps	Mean replans
No reservation; repulsion 8	518.2	96.92	34.0%	1,350.4
Reservation; repulsion 4	549.6	97.04	0.0%	1,232.5

Reservation at radius 4 was 6.1% slower, but improved AUC slightly and reduced both duplicate targeting and replanning. In the selected speed-first line, duplicate-target steps climbed from 19.7% on rooms to 55.8% on labyrinth, with corridors at 34.4% and bottlenecks at 26.0%. The ablation measures a real coordination trade-off: removing duplicated targets did not dominate the completion-speed objective.

5.4 Phase 4: EXMSG communication stress

All 1,360 Phase 4 jobs completed. None of the tested range, bandwidth, or loss settings produced a mission-failure threshold, so Table 8 reports illustrative degradation points instead. These are observed settings, not statistically estimated change points.

Table 8: Selected Phase 4 communication-stress results. Median steps are means of map-level medians. Bandwidth and loss rows use communication range 6.

Factor	Setting	Median steps	AUC	Outcome
Range	10 cells	467.4	97.42	Final speed-first setting
Range	8 cells	498.8	97.16	Near-fast lower-range reference
Range	3 cells	590.5	96.41	26.3% slower than range 10
Bandwidth	Unlimited	521.1	96.87	Unrestricted reference
Bandwidth	500 cells	525.9	96.86	Similar observed speed
Bandwidth	25 cells	626.6	96.41	20.2% slower than unlimited
Packet loss	15%	514.2	96.86	No completion loss observed

We kept communication range 10 for the final speed-first line, along with unlimited bandwidth, though a 500-cell cap gave almost the same observed map-level median. Packet loss up to 15% cost no completions inside the tested envelope. Since every setting completed, the campaign added no separate `exmsg_robust` finalist. The matched-tuning campaign later adopts this range-10

selection as a shared radio assumption: its local-communication unified line reuses it without retuning its own range, a limitation discussed in Section 6.5.

5.5 Phase 9 final validation: a reliability split and a reversed speed tier

All 3,150 Phase 9 jobs executed successfully across five lines: EXMSG (the 7E+3M finalist), tuned local unified at $N = 10$ and its equal-cost counterpart at $N = 8$, an idealized free-communication ceiling for unified (`unifiedComm=pool`, same tuned return policy and coordination), and EXANCH. The outcome splits in two. EXMSG and EXANCH complete every map. All three unified lines, the pool ceiling included, complete exactly 19 of 21, failing `labyrinth` and `labyrinth_big` at 0 of 30 seeds each. Unlimited, range-free communication does not rescue those two maps, so the failure is built into homogeneous return-to-hub exploration rather than into its radio. Among the 19 maps every line completes, though, tuned local unified at $N = 10$ is now faster than EXMSG in aggregate and wins most topology families outright: the reverse of the first campaign’s untuned comparison, where unified completed only 13 of 21 maps in the Phase 5 validation and was the slower line overall.

Table 9: Headline Phase 9 evidence. Step and AUC values are means of map-level medians (19 shared maps for steps, all 21 for AUC), with each map weighted equally. Pool and EXANCH are idealized ceilings, excluded from cost claims.

Metric	EXMSG	Uni. N10	Uni. N8	Uni. pool	EXANCH
Maps completed	21/21	19/21	19/21	19/21	21/21
Mean map-level median steps (19 shared)	310.3	299.5	322.1	291.5	521.6
Mean fixed-horizon coverage AUC (21)	98.70	95.55	95.42	95.67	97.67
Fleet-cost proxy	23.97	30.0	24.0	30.0	30.0
Communication attempts / 100 steps	761.7	1668.1	926.3	2001.6	520.4

On the 19 shared maps, tuned unified N10 is 3.5% faster than EXMSG in aggregate (299.5 against 310.3 mean map-level median steps), and EXMSG’s equal-cost twin, unified N8, is 3.8% slower than EXMSG at 322.1. The idealized pool ceiling gains only 2.7% over tuned local unified N10 (291.5 against 299.5), which says that idealized communication buys very little once return policy and coordination are tuned. EXMSG’s mean AUC of 98.70 beats all three unified variants (95.4–95.7) because AUC covers all 21 maps and unified’s two labyrinth failures drag it down; EXANCH’s 97.67 sits in between, reflecting its own slower completion on non-labyrinth maps despite finishing everything.

Table 10: Per-family speed and cost verdict. Steps are mean map-level medians on maps where the compared lines both complete (19 of 21 maps; labyrinth excluded from the unified columns because it never completes). r is the messenger cost ratio.

Family	EXMSG	Uni. N10	Uni. N8	Verdict
Labyrinth	734.7	fail	fail	EXMSG wins unconditionally; cost irrelevant
Corridors	670.2	895.3	948.0	EXMSG dominates speed and cost for $r \leq 1.0$ (vs. N10); trivially for $r \leq 1/3$ (vs. N8)
Bottlenecks	272.3	203.7	228.0	Unified faster at every tested r ; EXMSG's only lever is lower cost
Columns	152.5	113.3	135.3	Unified faster at every tested r
Loops	299.3	245.0	267.7	Unified faster at every tested r
Open	150.8	104.5	119.3	Unified faster at every tested r
Rooms	285.3	218.7	227.3	Unified faster at every tested r

Table 10 is the report's central result. Under matched tuning, EXMSG's speed advantage survives on exactly the two high-diameter, low-connectivity families: labyrinth, unconditionally, since unified never completes it, and corridors, where EXMSG dominates on both speed and cost for any tested messenger cost ratio r up to 1.0 against the equal-mobile-robot-count control, and trivially against the equal-cost control. On the other five families (bottlenecks, columns, loops, open, and rooms) tuned unified is faster at every tested r , and only cost still favors EXMSG. Figure 3 plots this directly: EXMSG's cost-vs-speed line sits below both unified points in the corridors and labyrinth panels, and above them everywhere else.

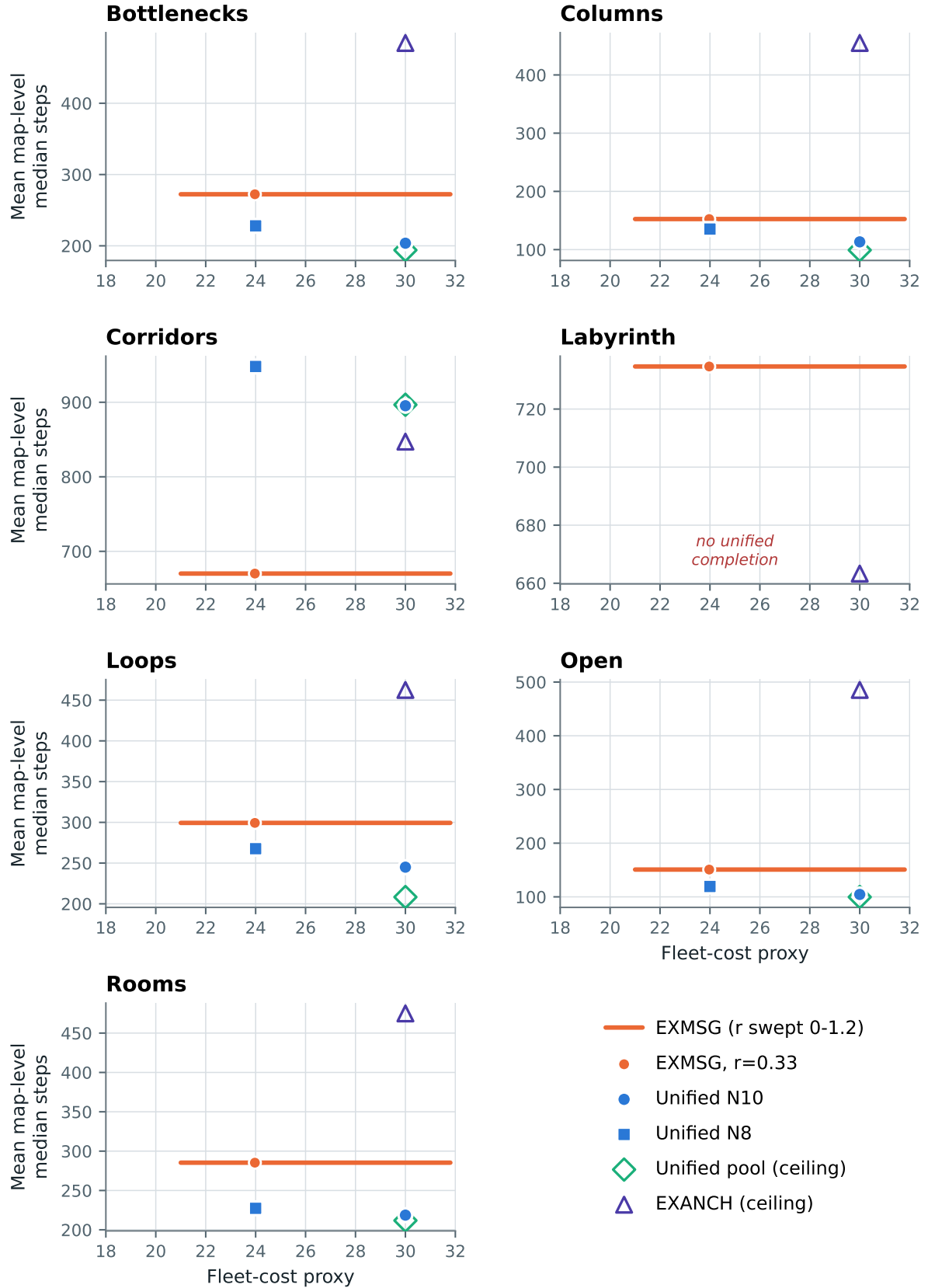


Figure 3: Fleet-cost proxy vs. mean map-level median completion steps, by map family, on the 19 maps every compared line completes. EXMSG's cost is a function of the messenger cost ratio r (swept 0 to 1.2) and is drawn as a line; unified N8/N10 and the idealized ceilings are fixed points. A line below both unified points is a speed-and-cost dominance region for EXMSG; a line above them means EXMSG's only lever left is cost.

Figure 4 shows the same reversal at the level of paired seeds instead of family means. EXMSG wins nearly every corridor and labyrinth seed pair against tuned unified N10, and loses nearly every bottleneck, column, loop, open, and rooms pair at standard and large scale. The pattern is not uniform across scale, however. Several families flip at small scale: `loops_small` is 30/0 to EXMSG while `loops_big` is 0/30 to unified, and `bottlenecks_small` is 26/4 to EXMSG while `bottlenecks_big` is 2/28 to unified. The diameter argument developed in Section 6 does not explain this small-vs-large reversal, and it remains an open question.

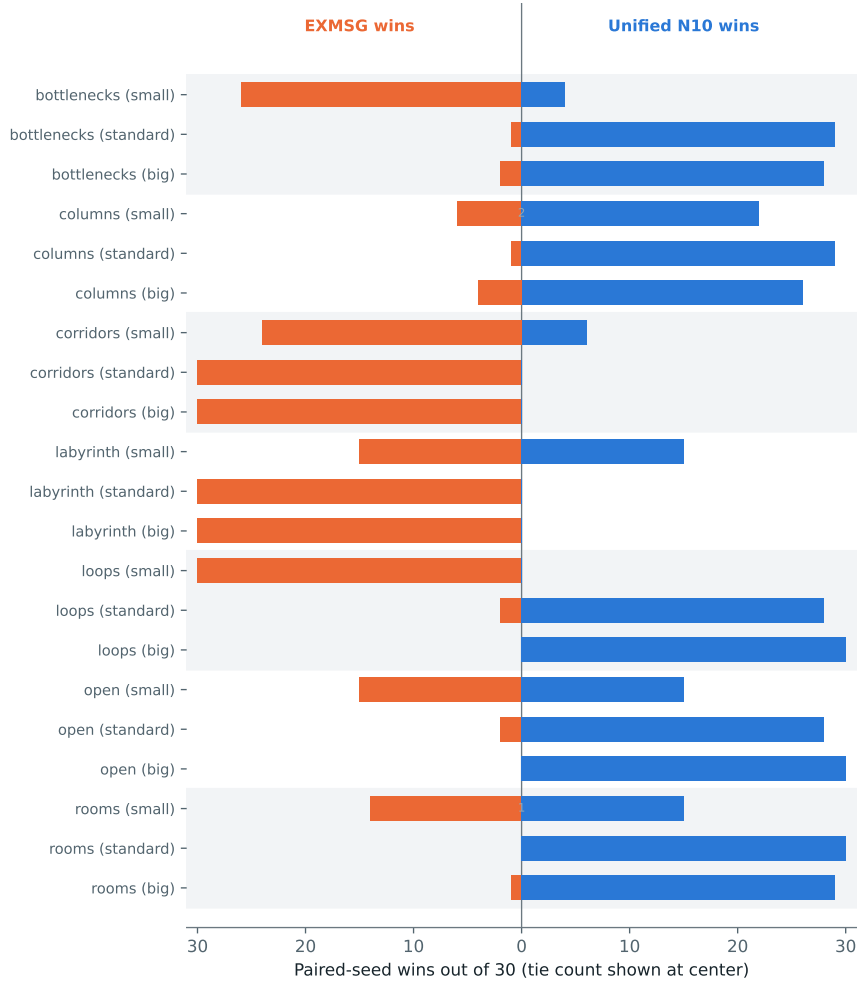


Figure 4: Paired-seed win counts (out of 30) for EXMSG vs. tuned local unified N10, by map and scale. A run wins on completion first, then on lower step count; the tie count is printed at the zero line. Families are grouped and lightly shaded for readability, ordered small/standard/large within each family.

Figure 5 puts both headline mechanisms in coverage-over-time terms: the persistent structural failure on `labyrinth_big`, where all three unified variants plateau below 75% coverage while EXMSG and EXANCH reach 100%, and the speed reversal on `rooms_big`, where tuned unified reaches full coverage well before EXMSG despite EXMSG’s own steady progress.

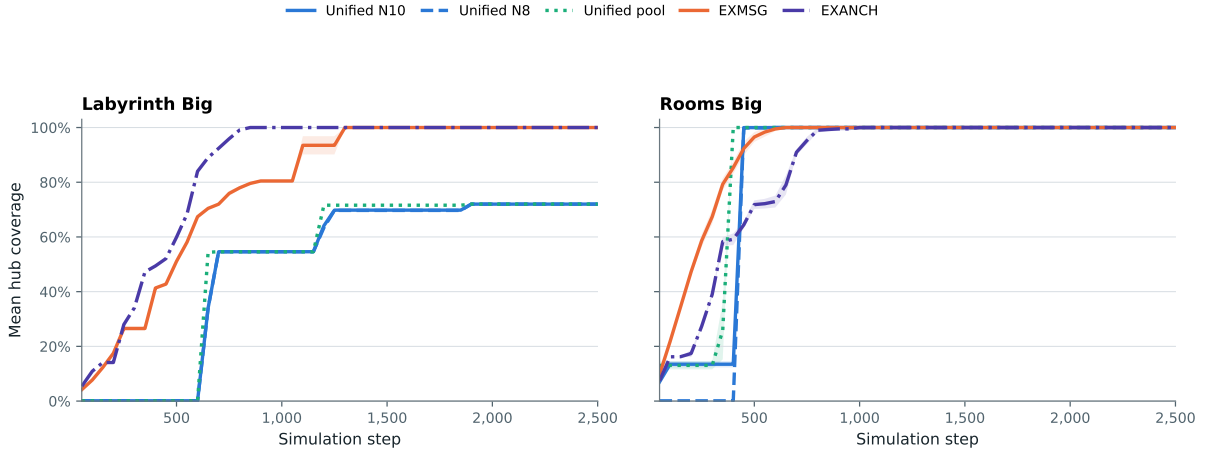


Figure 5: Final-validation hub-coverage curves on `labyrinth_big` (persistent structural failure for every unified variant, including the idealized pool ceiling) and `rooms_big` (a family where matched tuning reverses the speed ranking). Lines are means over 30 paired-seed runs sampled every 50 steps; terminal coverage is carried forward. The display stops at step 2,500 because no plotted line changes thereafter.

Taken together, these results argue for a topology-conditional reading of role specialization: it is what makes completion reliable on high-diameter, low-connectivity topologies, and elsewhere it trades speed for fleet cost. EXANCH and the pool ceiling sit outside that trade-off by construction. Both are idealized references, excluded from cost claims, useful only as upper bounds on what perfect relay infrastructure or perfect communication could deliver.

6 Discussion

6.1 When mobile specialization helps

The interpretation is an opportunity-cost argument, and matched tuning sharpens it from a blanket claim into a conditional one. In unified mode, the robot that sensed a frontier is also the robot that has to stop exploring and walk home. EXMSG hands that travel to messengers: in the matched-tuning campaign’s finalist, seven explorers keep sensing while three mobile relays do the delivering. Whether the trade pays depends on how expensive a return trip is relative to sensing. On high-diameter, low-connectivity topologies (corridors and labyrinth) a return trip is long, so paying robots to specialize in it and letting the rest sense uninterrupted wins outright (Table 10): EXMSG is both faster and cheaper there, and it is the only non-control architecture that completes labyrinth at all. On compact or well-connected topologies (bottlenecks, columns, loops, open, rooms) a return trip is cheap, and the sensing capacity EXMSG surrenders to its messengers is not worth what it buys: tuned unified is faster everywhere in this group, and EXMSG’s remaining advantage is its lower fleet-cost proxy. The mechanism agrees with the motivation for separating explorer and relay roles (de Hoog et al., 2009) and for separating data generation from upload under intermittent communication (Guo and Zavlanos, 2018), while narrowing where that motivation actually pays off empirically.

Matched tuning supports the conditional mechanism far better than the untuned comparison could, since the untuned comparison could not tell specialization apart from unified simply being undertuned. Once unified receives its own sequential tuning budget and the same communication constraint as EXMSG, the reliability gap persists unconditionally (Section 5.5) while the speed gap reverses on five of seven families. One coupled property beyond role structure still varies between the two lines, since EXMSG’s messengers are also priced below full explorer cost. What

the results support is therefore specialization as a topology-conditional architectural explanation, not a single-factor causal estimate.

6.2 Comparison with the literature

Table 1 sets seven useful alignments against their critical differences. The Phase 3 ablation is where the study first touches coordinated frontier exploration. Prior work discounts already assigned targets to limit redundant exploration (Burgard et al., 2000, 2005). Reservation hit that narrow goal in our simulator, driving duplicate-target steps to zero, and was still 6.1% slower than the repulsion-only line we selected. That does not contradict the literature; eliminating duplicate assignments and minimizing hub-delivery time are simply different objectives. What it shows is why coordination has to be judged at the mission level rather than by assignment cleanliness alone.

The hub objective is closest to work that folds communication into exploration feasibility or utility (Rooker and Birk, 2007; Pei et al., 2010). We relax continuous connectivity and measure the delivery delay that follows. Phase 4 found degradation at range 3 and at a 25-cell transfer cap without observing completion failure in the tested envelope, which fits the broader conclusion that connectivity and bandwidth shape exploration performance, even though the numerical boundaries are ours alone. Unlike CBAX, the experiment has no joint network-flow optimization, traffic model, or media stream; unlike continuous-connectivity formulations, it permits temporary disconnection.

EXMSG also resembles intermittent rendezvous and source-relay systems (Guo and Zavlanos, 2018; da Silva et al., 2024). In all three, information exchange is episodic rather than continuously available. What differs is how the meetings come about. The cited approaches reason about temporal tasks, buffers, or planned rendezvous; our messengers head for a heuristically selected explorer and then for the hub. The results therefore show that behavior-level specialization can work across the tested maps, not that it matches the guarantees or efficiency of formal rendezvous planning. Under matched tuning (Section 5.5) the data also show *where* this literature’s separation-of-roles argument holds: it wins when delivery travel is long relative to sensing travel and loses when it is short. The untuned comparison could not draw that line, because it never gave the homogeneous alternative a fair chance to be fast.

The stationary-relay comparison needs an even stronger qualification. Relay-placement work such as CARA reasons about the communication context of deployed support robots (Abu-Aisheh et al., 2022). EXANCH keeps the geometric idea of extending relay infrastructure and throws away the internal propagation cost through its common anchor pool. Its success on every map and its low communication-attempt count are best read as an optimistic reference for connected infrastructure. They are not evidence that a physical multi-hop mesh would show the same delay, bandwidth use, or reliability.

Finally, the project exchanges occupancy-grid cells, and that operation is not probabilistic mapping or multi-robot SLAM. Our zero-noise, common-frame setting omits the uncertainty updates of occupancy mapping (Elfes, 1989), the localization coupling of active mapping (Lluvia et al., 2021), and the alignment and consistency problems emphasized in multi-robot SLAM and map-merging reviews (Saeedi et al., 2016; Yu et al., 2020; Kegeleirs et al., 2021). The boundary matters: delivering perfectly aligned cells late is a far easier problem than merging uncertain maps from drifting coordinate frames.

6.3 Topology-dependent behavior

Table 10 and Figure 4 show where EXMSG’s delivery mechanism absorbs the least overhead once unified is tuned to the same standard. EXMSG wins labyrinth unconditionally and dominates corridors on speed and cost alike. A plausible reading is that mobile relays can shuttle through chokepoints and long passages repeatedly while the explorers stay beyond them, which separates delivery travel from sensing work. Phase 2’s delivery latency and delivered cells per step back up the weight of that travel burden: both degraded on corridors and labyrinth relative to rooms and bottlenecks. Unified, once given the same tuning budget and communication constraint as EXMSG, wins the other five families (bottlenecks, columns, loops, open, rooms) on speed at every tested messenger cost ratio. In those less constraining layouts the homogeneous fleet’s return-to-hub penalty is smaller than the sensing capacity EXMSG surrenders to its dedicated messengers, so the role split itself becomes the binding cost rather than delivery travel.

The paired-seed record complicates any purely diameter-based story. On **loops** and **bottlenecks**, EXMSG wins nearly every small-scale seed pair and loses nearly every large-scale one (Section 5.5, Figure 4). If mission diameter alone drove the mechanism, scale should not flip the winner inside a family whose topology is otherwise held fixed. The pattern is still open. The most specific candidate we can offer is that EXMSG’s delivery capacity is fixed while the map is not: three messengers serve a small map and a large one alike, so a carrier fraction chosen at standard scale has to cover several times the area without growing. Unified has no such ceiling. Its carrier fraction is endogenous, since every robot delivers its own map and any of them becomes a carrier the moment its return timer fires, up to the whole fleet at once, so delivery capacity scales with the same budget that produces the knowledge. On small maps the split EXMSG was given would then be enough, and its uninterrupted sensing wins; on large ones the binding constraint would be the split itself rather than the topology. Phase 8 fixed EXMSG’s composition on standard-scale maps only, so the campaigns carry no evidence either way, and testing it is left to future work (Section 7).

6.4 Trade-offs

No single scalar captures EXMSG’s position relative to unified and EXANCH, and matched tuning changes which metric comes first. Completion rate still precedes speed, because it is the one dimension on which the result is unconditional: EXMSG and EXANCH complete every map, while tuned unified fails exactly labyrinth and labyrinth_big no matter how its communication is idealized (Section 5.5). Once completion is equal, though, speed no longer favors EXMSG by default the way it did against the untuned reference. On five of seven families tuned unified is faster at every tested messenger cost ratio, and EXMSG’s fleet-cost proxy (23.97 against 24.00–30.00 for the unified variants) is its one remaining edge there. That proxy is itself a construction: messengers cost one third of an explorer and anchors cost zero, which is useful for evaluating the stated design budget and says nothing about real hardware, maintenance, or energy cost. Communication attempts are likewise simulator interactions specific to each architecture. Unified’s local variant counts range-limited pairwise exchanges the same way EXMSG does, its pool variant counts two common-pool interactions per explorer per step, and the anchor backbone requires no internal forwarding at all. The metric ranks activity under each implemented abstraction; it cannot be read as comparable radio packets or joules.

The coordination ablation and the Phase 6–7 return-policy and coordination tuning reveal the same multi-objective structure within and across architectures. Reservation gave EXMSG clean frontier ownership, fewer replans, and slightly higher AUC, while repulsion without reservation completed faster; Phase 7 independently landed on the same speed-first trade-off for unified. The communication stress study likewise found degradation before failure. Together these argue

for reporting a small set of outcomes per topology family (completion, speed, fixed-horizon coverage, communication activity, and cost proxy) instead of declaring a single winner, since a catalogue-wide aggregate hides exactly the reliability/cost split that Table 10 makes explicit.

6.5 Limitations and threats to validity

1. **Internal validity.** The matched-tuning campaign closes the main tuned-vs-untuned asymmetry: unified now has its own sequential tuning budget (return policy, then coordination) and the same range-limited communication EXMSG uses, so the EXMSG-vs-unified comparison differs only in role structure. Other asymmetries survive. EXANCH and the idealized free-communication unified ceiling are untuned by construction; both are reported as bounds and excluded from cost claims. Unified has no analogue of EXMSG’s communication-stress phase (Phase 4), since it has no bandwidth- or loss-bearing pairwise exchange between distinct roles to stress in the same way, so that dimension of the comparison stays EXMSG-specific. Communication range 10 was originally selected for EXMSG in Phase 4 and then reused as a shared radio assumption for the matched-tuning campaign’s local-communication unified line, without being retuned for unified. Phases 6 and 7 tune the return policy and coordination sequentially rather than jointly, so their separate contributions to the observed reversal are not isolated (see Future Work). The primary comparison still fixes $N = 10$; Phase 8’s cost sweep adds an $N = 8$ equal-cost control but does not establish full team-size scaling for either architecture.
2. **Construct validity.** Hub coverage is exact inside the simulator, but fleet cost is a chosen ratio and communication attempts are not packets, bytes, latency, or energy. Cell-count bandwidth also simplifies away message encoding, headers, routing, contention, and retransmission. Equal mobile-robot budgets do not mean equal total unit counts: EXANCH adds a map-dependent anchor inventory that the selected proxy prices at zero.
3. **Model validity.** Oracle navigation supplies ground-truth traversability; frontier candidates are filtered against ground truth; sensing is noise free; maps share one coordinate frame; unified’s pool variant uses a global explorer pool; and EXANCH uses an instantaneous common anchor pool. Every one of these removes a failure mode that physical systems have.
4. **External validity.** The 21 maps give controlled variation across seven designed grid families, not a random sample of buildings, disaster sites, or radio environments. Nothing here establishes real-robot performance.
5. **Statistical validity.** Step statistics for unified on labyrinth and labyrinth_big are truncated by the horizon and have to be read beside completion rate. The 30 paired seeds estimate stochastic variability within each fixed map, but the map catalogue is designed rather than sampled, so its 21 environments do not justify a population-level confidence interval. Normal-approximation intervals for per-map mean steps are descriptive, and neither campaign’s sequential tuning carries a statistical correction.

7 Future Work

The experiments worth running first are the ones that test the assumptions above:

1. realistic sensing and navigation: rerun the finalists under known-free or risk-unknown navigation and under sensing noise with belief mapping;
2. attribute the speed reversal: rerun Phases 6 and 7 in isolation, and add map instances to explain the small-vs-large reversal on `loops` and `bottlenecks` (Section 6);

3. scale the role ratio with the map: Phase 8 fixed EXMSG’s seven-explorer, three-messenger split on standard-scale maps only, so its delivery capacity stays constant while the area that capacity has to serve grows. Sweeping the split per map scale, against unified’s endogenous carrier fraction, would test the fixed-capacity hypothesis of Section 6;
4. a physical anchor network with hop-by-hop delay, loss, and bandwidth;
5. team-size scaling beyond the four core maps and the cost-frontier framing;
6. robot failures, stale information, localization drift, measured radio models, and hardware-based costs.

Two further directions extend the platform rather than test its assumptions. The simulator already keeps roles, frontier policies, navigation models, and communication behind configuration interfaces; documenting those as extension points would let others plug in their own behaviors and measure them against the recorded references. The hub-delivery objective is also a natural reward signal for reinforcement learning: classic or deep RL could learn frontier selection, messenger routing, or role allocation directly, with the hand-designed heuristics evaluated here serving as baselines.

8 Conclusion

We evaluated a specialized explorer–messenger architecture under a hub-delivered knowledge objective, first against a homogeneous return-to-hub design and an idealized stationary-relay design held as untuned references, then against the homogeneous design again once it had its own matched sequential tuning budget and the same communication constraint EXMSG uses. That second comparison is this report’s final result, and it splits into two tiers. EXMSG and the idealized relay design complete every one of 21 fixed maps. The retuned homogeneous design completes 19, failing the same two large maze-like maps every time, even in its own idealized free-communication variant, so that gap is structural rather than a tuning or bandwidth shortfall. Among the 19 maps every line completes, though, the retuned homogeneous design is now faster than EXMSG in aggregate (299.5 against 310.3 mean map-level median steps) and wins five of seven topology families outright. That reverses the untuned comparison, and it shows the first campaign had partly been measuring an undertuned baseline. EXMSG’s advantage survives unconditionally only on high-diameter, low-connectivity topologies: corridors, and labyrinth, where it is the only mobile-only line that completes at all. On the remaining five families it narrows to a lower fleet-cost proxy. The findings stay conditional on the tested fixed maps, oracle navigation, noise-free sensing, common map coordinates, and simplified communication, and on one open pattern, since several families reverse again between small and large scale in a way the diameter-based mechanism does not explain. Within those limits they support a topology-conditional interpretation rather than a universal opportunity-cost claim: specialized messengers absorb hub-delivery travel that is worth paying for only when that travel is itself expensive. Information-delivery architecture belongs in exploration design whenever useful mission knowledge must reach a base station, with role specialization chosen per topology rather than assumed.

References

- Razanne Abu-Aisheh, Francesco Bronzino, Lou Salaün, and Thomas Watteyne. CARA: Connectivity-aware relay algorithm for multi-robot expeditions. *Sensors*, 22(23):9042, 2022. doi: 10.3390/s22239042.

- Francesco Amigoni, Jacopo Banfi, and Nicola Basilico. Multirobot exploration of communication-restricted environments: A survey. *IEEE Intelligent Systems*, 32(6):48–57, 2017. doi: 10.1109/MIS.2017.4531226.
- Jan Bayer and Jan Faigl. Decentralized multi-robot exploration under low-bandwidth communications. *Autonomous Robots*, 50:7, 2026. doi: 10.1007/s10514-025-10234-3.
- Wolfram Burgard, Mark Moors, Dieter Fox, Reid Simmons, and Sebastian Thrun. Collaborative multi-robot exploration. In *Proceedings of the IEEE International Conference on Robotics and Automation*, volume 1, pages 476–481, 2000. doi: 10.1109/ROBOT.2000.844100.
- Wolfram Burgard, Mark Moors, Cyrill Stachniss, and Frank E. Schneider. Coordinated multi-robot exploration. *IEEE Transactions on Robotics*, 21(3):376–386, 2005. doi: 10.1109/TRO.2004.839232.
- Alysson Ribeiro da Silva, Luiz Chaimowicz, Thales Costa Silva, and M. Ani Hsieh. Communication-constrained multi-robot exploration with intermittent rendezvous. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 3490–3497, 2024. doi: 10.1109/IROS58592.2024.10802343.
- Julian de Hoog, Stephen Cameron, and Arnoud Visser. Role-based autonomous multi-robot exploration. In *Proceedings of ComputationWorld 2009*, pages 482–487, 2009. doi: 10.1109/ComputationWorld.2009.14.
- Alberto Elfes. Using occupancy grids for mobile robot perception and navigation. *Computer*, 22(6):46–57, 1989. doi: 10.1109/2.30720.
- Meng Guo and Michael M. Zavlanos. Multi-robot data gathering under buffer constraints and intermittent communication. *IEEE Transactions on Robotics*, 34(4):1082–1097, 2018. doi: 10.1109/TRO.2018.2830370.
- Miquel Kegeleirs, Giorgio Grisetti, and Mauro Birattari. Swarm SLAM: Challenges and perspectives. *Frontiers in Robotics and AI*, 8:618268, 2021. doi: 10.3389/frobt.2021.618268.
- Iker Lluvia, Elena Lazkano, and Ander Ansuategi. Active mapping and robot exploration: A survey. *Sensors*, 21(7):2445, 2021. doi: 10.3390/s21072445.
- Yuanteng Pei, Matt W. Mutka, and Ning Xi. Coordinated multi-robot real-time exploration with connectivity and bandwidth awareness. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 5460–5465, 2010. doi: 10.1109/ROBOT.2010.5509803.
- Martijn N. Rooker and Andreas Birk. Multi-robot exploration under the constraints of wireless networking. *Control Engineering Practice*, 15(4):435–445, 2007. doi: 10.1016/j.conengprac.2006.08.007.
- Sajad Saeedi, Michael Trentini, Mae Seto, and Howard Li. Multiple-robot simultaneous localization and mapping: A review. *Journal of Field Robotics*, 33(1):3–46, 2016. doi: 10.1002/rob.21620.
- Brian Yamauchi. A frontier-based approach for autonomous exploration. In *Proceedings of the IEEE International Symposium on Computational Intelligence in Robotics and Automation*, pages 146–151, 1997. doi: 10.1109/CIRA.1997.613851.
- Brian Yamauchi. Frontier-based exploration using multiple robots. In *Proceedings of the Second International Conference on Autonomous Agents*, pages 47–53, 1998. doi: 10.1145/280765.280773.

Shuien Yu, Chunyun Fu, Amirali K. Gostar, and Minghui Hu. A review on map-merging methods for typical map types in multiple-ground-robot SLAM solutions. *Sensors*, 20(23):6988, 2020. doi: 10.3390/s20236988.

A The Simulator

The experimental artifact behind both campaigns is a complete, reusable simulator. A single Python engine implements the world model, the three architectures, sensing, communication, and every metric. Every result here, from interactive inspection through tuning sweeps to final validation, runs through that one engine, so the two views of the system can never disagree.

Interactive front end. A FastAPI server exposes the engine over WebSocket to a React browser interface with live map canvases, two side-by-side simulation slots, and control panels for every parameter of Appendix B. Only changed cells go over the wire each frame, which keeps large maps responsive. This is the interface that carried the plateau diagnostic preceding Phase 6.

Headless campaign runner. Batch experiments run without the server. A campaign is a JSON manifest declaring base parameters, parameter grids, seed blocks, horizons, and repeat counts. The runner expands that manifest into jobs, executes them in parallel worker processes, and writes a resolved copy of the manifest, environment metadata, one CSV row per run, and grouped summary statistics: completion rate, mean and median steps, population standard deviation, normal-approximation confidence intervals, and coverage curves. A separate interpretation script applies the ranking rule of Section 4.

Performance. Maps and knowledge grids are NumPy arrays, and the per-step hot paths (frontier tests, line-of-sight, BFS fields) are JIT-compiled with Numba. Each simulation owns its own seeded random-number generator, so concurrent interface slots and parallel batch workers stay independently reproducible.

Reproducibility. Seed blocks start at 260518100–260518500 for Phases 1–5 and at 260518050 (preflight), 260518600, 260518800, 260518900, and 260519000 for Phases 6–9; every combination within a phase uses paired seeds. Each phase archive keeps its resolved manifest, per-run CSV, and grouped summary, and the matched-tuning campaign additionally logs one dated decision entry per phase recording the selected configuration, the ranking rule, and the runner-up gap. The figures of Section 5.5 are regenerated straight from the archived Phase 9 run records by scripts kept beside the report source.

B Tunable Parameter Catalogue

Tables 11 and 12 list every parameter of the simulation interface with its default value, named as it appears in the campaign manifests and the WebSocket API. Any configuration from either campaign is fully described by this catalogue plus the finalist deviations listed at the end of the appendix.

Table 11: Parameter catalogue I: mission, team, and sensing.

Parameter	Default	Meaning
<code>mode</code>	<code>exmsg</code>	Architecture: <code>unified</code> , <code>exmsg</code> , or <code>exanch</code>
<code>mapKey</code>	<code>rooms</code>	Fixed map to load; <code>random</code> generates one procedurally
<code>gridSize</code>	40	Side length of procedurally generated maps
<code>obstacleDensity</code>	0.20	Obstacle fraction of procedurally generated maps
<code>seed</code>	42	Master random seed for the run
<code>robotCount</code>	10	Mobile-robot budget N
<code>explorerRatio</code>	0.6	Explorer share of N in EXMSG; the remainder become messengers
<code>messengerSpeed</code>	2	Messenger grid moves per simulation step
<code>messengerPolicy</code>	<code>nearest</code>	Messenger target choice: nearest explorer or expected knowledge gain
<code>hubReturnInterval</code>	40	Unified: steps before the hub-return timer becomes eligible
<code>hubReturnIdle</code>	10	Unified: idle steps required after the timer fires
<code>hubReturnMax</code>	90	Unified: hard cap that forces a return regardless of activity
<code>unifiedComm</code>	<code>pool</code>	Unified sharing: global team map (<code>pool</code>) or range-limited exchange (<code>local</code>)
<code>senseRange</code>	5	Sensing radius in cells
<code>coneAngle</code>	120	Sensing cone width in degrees
<code>falseEmptyRate</code>	0	Probability a sensed obstacle reads as free
<code>falseObstacleRate</code>	0	Probability a sensed free cell reads as an obstacle
<code>noiseModel</code>	<code>locked</code>	First observation wins (<code>locked</code>) or vote-based revision (<code>belief</code>)

Table 12: Parameter catalogue II: coordination, navigation, communication, cost, and failures.

Parameter	Default	Meaning
<code>frontierPolicy</code>	<code>nearest</code>	Frontier selection strategy (ten available)
<code>frontierReservation</code>	<code>false</code>	Exclusive frontier claims with 30-step expiry
<code>useRepulsion</code>	<code>true</code>	Penalize frontiers near teammates' targets
<code>repulsionRadius</code>	4	Radius of the repulsion penalty
<code>navigationModel</code>	<code>oracle</code>	Motion checks against ground truth, known-free cells, or risk-weighted unknowns
<code>commRange</code>	6	Robot-to-robot communication range in cells
<code>anchorCommRange</code>	6	Explorer-to-anchor range (EXANCH)
<code>anchorDensity</code>	0.015	Planned anchors per free cell (EXANCH)
<code>commBandwidthCells</code>	0	Maximum cells per transfer; 0 means unlimited
<code>packetLossRate</code>	0	Probability an eligible transfer is dropped
<code>linkQualityModel</code>	<code>none</code>	Optional distance- or line-of-sight-dependent link quality
<code>cellPriority</code>	<code>newest</code>	Which cells to send first under a bandwidth limit
<code>explorerCost</code>	3.0	Fleet-cost proxy per explorer
<code>messengerCostRatio</code>	0.33	Messenger cost as a fraction of explorer cost
<code>anchorCostRatio</code>	0.0	Anchor cost as a fraction of explorer cost
<code>failureStep</code>	<code>none</code>	Step at which scheduled robot failures occur
<code>robotFailureRate</code>	0	Probability each robot fails at that step
<code>speed</code>	20	Live-interface steps per second; no effect on recorded results

The Phase 9 finalists depart from these defaults only as follows. EXMSG: `explorerRatio` 0.7, `messengerSpeed` 4, `commRange` 10, and `repulsionRadius` 8. Tuned local unified: `unifiedComm` `local`, `hubReturnMax` 360, `commRange` 10, `repulsionRadius` 8, and `robotCount` 10 or 8. The pool ceiling takes the same unified values with `unifiedComm` `pool`. EXANCH runs entirely on defaults. Every line shares sensing range 5, cone 120°, oracle navigation, zero noise, unlimited bandwidth, and zero packet loss. The complete per-line configuration is stored in the archived Phase 9 manifest.